



Branch/Semester	B.Tech CSE (AI & ML) / Semester V
Subject Name:	Design and Analysis of Algorithms
Subject Code:	01AI0506
Assignment:	Assignment No. I (Result Improvement Programme)
Date:	7 th September 2026
Faculty Name:	Prof. Abhishek Chauhan

Sr.	Algorithm Design and Analysis: Complexity Analysis	CO	BL
1	Apply Euclid's algorithm to find the GCD of two given numbers. Represent it using both pseudocode and step-by-step description, and demonstrate with a suitable example.	CO1	BL3
2	Apply the characteristics of an algorithm to develop a step-by-step solution for finding the largest among three numbers.	CO1	BL3
3	Analyze the time complexity of an algorithm containing single and nested loops I. void example(int n) { while (n > 1) { n = n / 2; } } II. void example(int n) { for (int i = 1; i <= n; i++) { printf("%d ", i); } printf("Done"); }	CO2	BL4

	<pre> III. void example(int n) { int x = 0; for (int i = 1; i <= n; i++) { x++; } for (int i = 1; i <= n; i++) { for (int j = 1; j <= n; j++) { x++; } } } </pre>		
4	Analyze the difference between time efficiency and space efficiency of an algorithm.	CO2	BL4
5	Analyze the concept of asymptotic analysis and discuss its importance in evaluating and comparing algorithm performance as input size increases.	CO3	BL4
6	Evaluate the differences among Big-O, Big-Omega, and Big-Theta notations, and justify their significance in determining upper, lower, and tight bounds using suitable examples.	CO3	BL5
7	Apply Big-O, Big-Ω, and Big-Θ notations to analyze: (i) $f(n) = 3n^2 + 5n + 10$, (ii) $f(n) = 7n^3 + 4n + 8$.	CO3	BL3
8	Analyze the growth rates of n , n^2 , n^3 , and 2^n . Compare their rates of growth and explain how they affect algorithm efficiency.	CO3	BL4
9	Solve the recurrence relation $T(n) = 2T(n/2) + n$. Show the necessary steps and determine its asymptotic time complexity in Big-O notation.	CO2	BL3
10	Solve the recurrence $T(n) = 3T(n/2) + n$ using the Master Theorem. Identify the appropriate case and obtain the asymptotic complexity.	CO2	BL3